Series Worksheet

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely.

1. $\displaystyle\sum_{k=1}^{\infty} \frac{\cos(3k)}{k^2}$

2. $\displaystyle\sum_{k=1}^{\infty} \frac{k-1}{k^{3/2}}$

3. $\displaystyle\sum_{k=3}^{\infty} \frac{4}{k\sqrt{\ln(k)}}$

4. $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^k 7^{2k}}{(2k)!}$

5. $\displaystyle\sum_{k=1}^{\infty} \sqrt[k]{k}$

6. $\displaystyle\sum_{k=1}^{\infty} \frac{\ln(k^2)}{k}$

7. State and explain the Zero-Infinity Comparison Test